

■ NiyamPatch: Policy-to-Code Operational Cockpit

ChatGPT Codex India Hackathon 2026 — Project Description & Technical Documentation

Project Name	NiyamPatch: Policy-to-Code Operational Cockpit
Hackathon Track	AI & Governance / Autonomous Coding Systems (ChatGPT Codex 2026)
Repository URL	https://github.com/esafshanum/Niyam-Patch
Live Local URL	http://localhost:3000 (Next.js 16 App Router)
Seeded Scenario	MP Higher Education Scholarship Policy (Gazette #HE/2026/302)
Design Palette	Pastel Baby Blue (#E0F2FE) & Snow White (#FFFFFF)

1. Executive Summary

NiyamPatch is an enterprise-grade Policy-to-Code Operational Cockpit engineered to eliminate the multi-month implementation lag, manual human translation errors, and compliance risks involved in converting newly published government gazette circulars into production software logic.

When state governments amend public welfare policies (e.g., expanding eligibility income ceilings, revising scholarship criteria, or updating tax slabs), software portals typically lag by months. This lag leads to eligible citizens being wrongfully rejected by outdated code rules.

NiyamPatch solves this bottleneck by establishing an automated, Agentic Codex Pipeline that ingests gazette circular PDFs, extracts structured rule changes with 99.4% confidence, maps them directly to codebase AST constants (MAX_FAMILY_INCOME), executes 5 boundary unit tests, and presents color-coded git diffs for mandatory Human Administrator approval before applying patches.

2. Problem Statement

- 1. Massive Implementation Lag (3–6 Months):** After a policy is officially gazetted, IT vendors require months to write change requests, draft specifications, update codebases, and redeploy production servers.
- 2. Citizen Denial & Benefit Loss:** During the implementation lag, citizens meeting newly gazetted criteria (e.g. family income between ■2.5L and ■3.0L) are automatically rejected by legacy code engines.
- 3. Absence of Safety Controls in Generic AI:** Unchecked LLM code generation risks hallucinating critical parameters or breaking surrounding production logic, leading to catastrophic governance failures.
- 4. Lack of Auditable Legal Chain of Custody:** Government auditors (CAG/NIC) lack transparent traceability linking a specific code constant modification back to an official gazette circular clause number and approving officer.

3. Key Feature Matrix

Feature Module	Capabilities & Implementation
1. Security Sign-In Gate	Role-based landing screen with 1-click Quick Demo Sign-In.
2. Policy PDF Reader	Interactive reader for gazette orders with page navigation & text highlights.

3. Indic Vernacular Engine	1-click full-project translation across English, Hindi (■■■■■■■), and Marathi (■■■■■■■).
4. Voice Policy Assistant	Text-to-speech audio reader speaking circular summaries aloud in 3 Indic languages.
5. Codex Thought Stream	Collapsible terminal log drawer displaying step-by-step agentic AI reasoning.
6. AST Code Matcher	Side-by-side git diff viewer mapping Clause 4.1 to MAX_FAMILY_INCOME constant.
7. Boundary Test Matrix	5 in-memory boundary unit tests verifying edge case thresholds (■2.499L to ■3.00001L).
8. Human Safety Gate	Zero unverified code execution; requires explicit sign-off by Dr. A. K. Varma.
9. Split-Screen Simulator	Live comparison demonstrating applicant status on Legacy vs NiyamPatch Live engine.
10. Conflict Detector	Cross-checks circulars against 12 historical state acts to verify legal consistency.

4. Executable Boundary Test Suite Results

Test ID	Income Value	Boundary Condition	Pre-Patch Result	Post-Patch Result
TEST_1	■2,49,999	Sub-Boundary	PASSED (Eligible)	PASSED (Eligible)
TEST_2	■2,50,000	Baseline Ceiling	PASSED (Eligible)	PASSED (Eligible)
TEST_3	■2,80,000	Revised Mid-Tier	FAILED (Rejected ■)	PASSED (Eligible ■)
TEST_4	■3,00,000	New Gazette Ceiling	FAILED (Rejected ■)	PASSED (Eligible ■)
TEST_5	■3,00,001	Super-Boundary	PASSED (Rejected ■)	PASSED (Rejected ■)

5. Conclusion & Hackathon Submission Links

NiyamPatch successfully proves that agentic AI, combined with precise AST code matching and strict human safety gates, can transform government policy operations from a multi-month delay into a 60-second verified deployment.

GitHub Repository: <https://github.com/esafshanum/Niyam-Patch>

Local Development Server: <http://localhost:3000> (npm run dev)